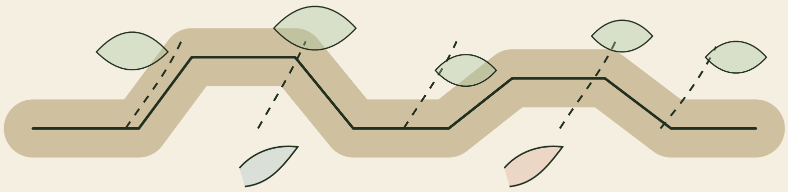
09 SECTIONS & DIAGRAMS



nabta WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY · AI PARK DESIGN CHALLENGE · DRAFT COMPETITION ENTRY

D·09

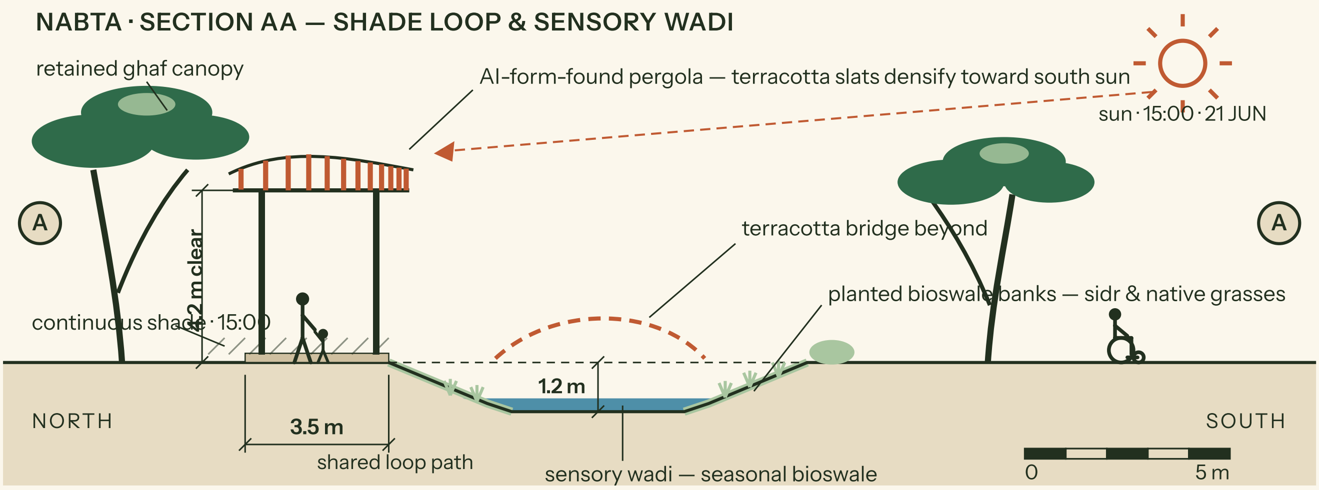
Concept Diagrams, Sections & Elevations

№10 Diagrams | №11 Sections

Package 9 — Concept Diagrams, Preliminary Design Drawings, Sections & Elevations

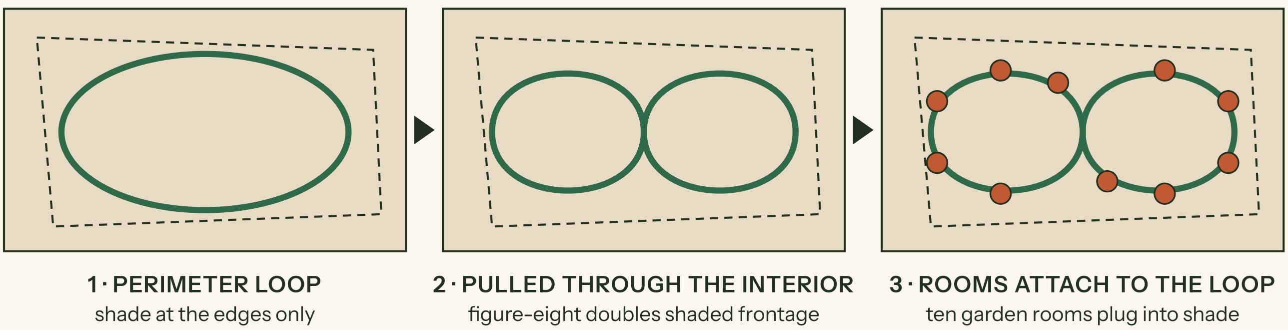
`Deliverable 10` `Deliverable 11` This package presents the graphic core of the «نبتة» **NABTA — Wellness Park** entry for Al Safa 2 (15,001 m² per the official as-built survey), under the concept **The Living Shade Loop**: the primary illustrative section through the shade loop and sensory wadi, the three-step concept evolution diagram, and the register of preliminary design drawings issued with the full submission. All drawings derive from a single federated CAD/BIM model, as required by the brief; the section below is reproduced at reading scale from drawing L-201.

SECTION AA — THROUGH THE SHADE LOOP AND SENSORY WADI (1:200 @ A2, REPRODUCED AT READING SCALE)



L-201 · SECTION AA. THE 3.5 M SHARED LOOP PATH RUNS ALONG THE NORTH BANK OF THE WADI UNDER A 4.2 M-CLEAR PERGOLA WHOSE TERRACOTTA SLATS DENSIFY TOWARD THE SOUTHERN SUN, JOINING THE RETAINED GHAF CANOPY TO HOLD CONTINUOUS SHADE ACROSS THE PATH AT THIS SECTION AT 15:00 ON THE SUMMER SOLSTICE — 92% COVER ACROSS THE FULL LOOP, VERIFIED 91.6% IN SIMULATION. THE 1.2 M-DEEP BIOSWALE DETAINS STORMWATER AND RECYCLED IRRIGATION FLOWS BETWEEN PLANTED SIDR BANKS; A TERRACOTTA BRIDGE CROSSES BEYOND THE CUT LINE.

CONCEPT EVOLUTION — FROM RING TO LIVING LOOP



L-010 (EXTRACT). A PERIMETER RING ALONE SHADES ONLY THE EDGES; PULLING A SECOND CIRCUIT THROUGH THE INTERIOR AS A 398 M FIGURE-EIGHT DOUBLES SHADED FRONTAGE AND CROSSES THE WADI TWICE. TOGETHER WITH THE 460 M PERIMETER RING IT FORMS AN 858 M CONTINUOUS SHADED NETWORK — ROUGHLY 1 KM INCLUDING THE WADI SPUR. THE TEN GARDEN ROOMS THEN ATTACH DIRECTLY TO THE LOOP, SO EVERY DESTINATION SITS WITHIN THE SHADE SYSTEM.

DRAWING REGISTER — PRELIMINARY DESIGN SET

All drawings are cut from one federated Revit/Rhino model set out on the official as-built parcel — an irregular 16-vertex polygon of 15,001 m² within a ~120 × 178 m envelope, street edge to the south, site grid rotated ~47° from north — with shared site coordinates, IFC 4 exchange and ISO 19650 naming, issued as DWG + PDF per the brief's CAD/BIM requirement. Sections and elevations reference the common ±0.000 park datum shown dashed in Section AA.

NO.	TITLE	SCALE	FORMAT
L-001	Illustrative masterplan	1:500	A0 · DWG/PDF from BIM
L-010	Concept & systems diagrams (shade, water, sensing)	NTS	A3 · PDF
L-101	Shade loop & garden rooms — general arrangement	1:250	A1 · DWG
L-201	Section AA — shade loop & sensory wadi	1:200	A2 · DWG/PDF
L-202	Section BB — play dune, majlis plaza & café terrace	1:200	A2 · DWG
L-301	Pergola elevations, north & south faces	1:100	A1 · DWG from Revit (LOD 300)
L-302	Majlis plaza & café terrace elevations	1:100	A1 · DWG
L-401	Pergola node & slat-fixing details	1:20	A3 · DWG
L-402	Bioswale edge, weir & terracotta bridge details	1:20	A3 · DWG
L-501	Planting & irrigation zoning plan	1:250	A1 · DWG + GIS
L-601	AI form-finding record — inputs, objectives, selected generation	NTS	A3 · PDF + parametric definition

HOW THE PERGOLA WAS FORM-FOUND

The pergola geometry in Section AA is not drawn — it is computed, then judged by the design team. A notional canopy surface over the loop centreline was meshed and exposed to an annual solar simulation at the Al Safa 2 coordinates. The mapping is direct: **simulated solar load per mesh face → slat density**. The most irradiated faces receive slats at 180 mm centres, tapering linearly to 600 mm centres on the sheltered north edge — the south-densifying gradient visible in the section.

An evolutionary solver then optimised slat spacing, chord curvature and post positions against a single primary target — **shade along the 3.5 m path centreline at 15:00 on the summer solstice**, the governing heat-stress hour drawn in Section AA, where the selected geometry reaches 92% cover (verified 91.6%) — with minimum slat material and preserved low-angle December sun as secondary objectives. The team reviewed the leading generations against the 4.2 m clear height and sightlines to the 1.2 m wadi before selecting one; that decision, and the solver record, are documented on sheet L-601. The selected generation exports as native parametric geometry alongside the IFC model, so the same definition regenerates fabrication data at detailed design. The result is a roof that visibly "reads" the sun — denser where Dubai's sky is harshest, open where the retained ghaf canopy already does the work.